Techfest 2026-27

InnovateX

InnovateX: The Future Habitats Challenge is a premier national innovation competition calling on India's youngest minds and future leaders to solve critical urban and environmental challenges for a sustainable tomorrow. Hosted by Techfest, IIT Bombay, Asia's Largest Science & Technology Festival, this challenge is a platform for building practical, high-impact solutions from the ground up.

We are seeking young problem-solvers, creators, designers, and systems thinkers who want to move beyond textbook concepts and create tangible prototypes that can transform our cities, starting from their own neighborhoods and scaling up to the national stage.

Competition Themes

Theme 1 : Underground Water Management

Domain Overview

IoT-based Underground Water Management focuses on using connected sensors, monitoring systems, and data analytics to understand and preserve groundwater resources. Cities face severe challenges in groundwater depletion, contamination, illegal extraction, pipeline leakages, and unpredictable recharge rates. IoT technologies can provide real-time insights into water levels, flow behaviour, and quality parameters, enabling smarter and more sustainable water management. Effective solutions in this domain contribute directly to sustainability and resource security.

Challenge Question

How can IoT technologies be designed and deployed to effectively monitor, protect, and manage underground water resources in an urban city? Propose an innovative IoT-based solution that addresses a specific groundwater-related challenge such as water level monitoring, contamination detection, illegal

extraction prevention, pipeline leakage detection, or aquifer recharge analysis. Your solution should detail the problem, proposed technology, impact, and feasibility within a framework.

Objectives and Deliverables

• **A Specific Underground Water Problem**: Participants should identify a clear challenge related to groundwater levels, quality, usage, or distribution.

• **Propose an IoT-Based Solution:** Present a comprehensive solution using IoT sensors, communication systems, and analytics tools, highlighting innovation, scalability, and practicality.

• **Demonstrate Technical Feasibility**: Explain how the IoT system would function, including sensor operation, data collection, network requirements, and real-time monitoring features.

• **Identify Operational Challenges**: Address issues such as underground connectivity, sensor durability, power consumption, maintenance, and data reliability.

• **Economic Analysis**: Evaluate the cost-effectiveness of the proposed IoT solution and suggest a sustainable model for large-scale adoption in cities.

• **Environmental and Social Impact**: Assess how the solution supports groundwater conservation, reduces wastage, improves water quality, and benefits communities.

Theme 2: Rising AQI Levels in Modern Urban Cities

Domain Overview

Air pollution has become one of the most critical environmental challenges in rapidly urbanizing cities. Increasing vehicular emissions, industrial activity, construction dust, and waste burning contribute to escalating Air Quality Index (AQI) levels. Prolonged exposure to poor air quality leads to respiratory illnesses, reduced visibility, and long-term environ- mental degradation. As cities continue to grow, smart-city technologies must play a vital role in monitoring, predicting, and reducing AQI to ensure healthier and more sustainable urban living.

Challenge Question

How can technologies be leveraged to monitor, predict, and reduce rising AQI levels in modern urban environments? Propose an innovative solution using IoT sensors, AI-driven air quality forecasting, adaptive traffic control, drone-based monitoring, or urban planning strategies. Your solution should address a specific pollution source and outline its benefits, feasibility, and implementation strategy

Objectives and Deliverables

• **Address a specific problem**: Participants should identify a primary contributor to rising AQI-vehicular emissions, industrial zones, construction dust, waste burning, etc.

• **Propose an Innovative Solution**: Develop a feasible, sustainable, and scalable solution using smart technologies to reduce or control air pollution.

• **Demonstrate Practical Feasibility**: Explain how the proposed system operates and justify its effectiveness in real-world scenarios.

• **Implementation Plan**: Provide a deployment strategy, including sensor placement, data collection, analytics, and integration with existing city systems.

• **Identify Potential Challenges**: Discuss obstacles such as sensor calibration, weather effects, maintenance, or system reliability.

• **Analyze Economic Viability**: Present a cost-effective plan for large-scale adoption and long-term operation in urban regions.

• **Evaluate Environmental Impact**: Quantify how the solution contributes to pollution reduction, public health improvement, and sustainability.

Theme 3: Rising plastic pollution in Coastal Cities

Domain Overview

Coastal plastic pollution has become one of the most critical environmental challenges in rapidly urbanizing megacities. In a coastal hub like Mumbai,

increasing single-use plastic consumption, unsegregated municipal waste, open dumping, and stormwater runoffs contribute to escalating marine debris. Survey data estimates that approximately **5 million kilograms (5,000 metric tonnes) of plastic waste** flow annually from Mumbai's waterways, creeks, and stormwater drains (*nullahs*) into the Arabian Sea. Prolonged accumulation of this waste chokes critical mangrove ecosystems, severely disrupts marine life, and worsens urban flooding during monsoons. As coastal cities continue to grow, smart-city technologies, automated interception systems, and circular-economy models must play a vital role in monitoring, capturing, and upcycling plastic waste to ensure cleaner and more sustainable urban-marine living.

Challenge Question

How can technologies be leveraged to monitor, intercept, and reduce rising plastic waste leakage into Mumbai's coastal and urban drainage networks? Propose an innovative solution using IoT waste-level sensors, automated river/creek skimmers, AI-driven sorting mechanisms, barrier-net systems, or community-integrated recycling strategies. Your solution should address a specific plastic waste source and outline its benefits, feasibility, and implementation strategy.

Objectives and Deliverables

- **Address a Specific Problem:** Participants should identify a primary contributor to Mumbai's marine plastic pollution such as single-use plastics entering stormwater drains, creek-mouth accumulation, unsegregated household garbage, or beach litter.
- **Propose an Innovative Solution:** Develop a feasible, sustainable, and scalable hardware or software solution using smart technologies to intercept, clear, or upcycle waste effectively.
- **Demonstrate Practical Feasibility:** Explain how the proposed system operates in real-world coastal or urban environments and justify its mechanical or digital effectiveness.
- **Implementation Plan:** Provide a deployment strategy, including installation sites (e.g., specific creeks, drainage outlets like Malad or Trombay, or high-footfall coastal zones), data collection, analytics, and integration with existing municipal systems.
- **Identify Potential Challenges:** Discuss obstacles such as marine corrosion, tidal fluctuations, regular maintenance, monsoon overflow, or system reliability.

- **Analyze Economic Viability:** Present a cost-effective plan for large-scale adoption, maintenance, and long-term operation within urban regions.
- **Evaluate Environmental Impact:** Quantify how the solution contributes to plastic reduction, marine life preservation, urban flood prevention, and overall sustainability.

Participants are allowed to choose any one of the Theme from the above themes

Competition Structure & Timeline

Eligibility & Team Rules

- **Team Size:** A maximum of 4 members per team.
- **Eligibility:** Open to all students with a valid ID from their educational institution. Cross-institutional teams are encouraged.

Competition Format

Round 1: Abstract Submission & Presentation

Round 1 requires teams to pitch their concept physically at their regional zonal center, showcasing their proposed architecture, design flow, and feasibility plan.

Your submission should be clear, concise, and original, containing:

- Team Information – Team name, Team ID,Theme number and contact details for all members.
- Problem Statement – A precise description of the specific problem you are addressing from one of the 3 competition themes.
- Background & Context – Briefly mention existing solutions (if any) and why they fall short.
- Proposed Solution – A short explanation of your original and innovative idea, including the technology you plan to use and how it works.
- Impact & Scalability – A brief summary of the potential social and economic impact and how it can be scaled to benefit a large population.

Note : Each team will be getting 10 Mins of time to present their solution.

The **Top five** people from each regional zonal center will qualify for Round 2

All the students who qualified for round 2 , should start working on their prototype.

Note : You do not need to make any submissions for Round 1; all submissions are evaluated at the zonal venues.

Round 2: Final Report & Prototype Submission and Final Presentations
Submit a detailed report and working prototype including:

- Detailed Solution Description – How it works, its design, and technology used.
- Prototype Demonstration – Evidence of functionality and usability for target users
- Cost & Market Viability – Estimated cost, affordability, and potential market fit.
- Implementation Plan – Steps for deployment and adaptability to different conditions.
- Scalability Plan – How it can be expanded for wider impact across India.
- Impact Assessment – Clear estimation of social and economic benefits.

The final report must be submitted by December 1, 2026 on the website.

Submission Guidelines:
All online submissions for this competition must be made through the official Techfest website prior to the specified deadline. Only the Team Leader is authorized to submit on behalf of the team. Submissions made by any other team member will not be considered valid, multiple submissions from the same team will not be accepted and may lead to disqualification from the competition.

Final Presentations

Final presentations will be held at Techfest, IIT Bombay, from **December 15-18, 2026**. This is the stage where the top teams will compete for the final prize.Please note that the decisions of the organizers and judges are final and binding. Participants must submit their Abstract Submission and Final Report on the website.

Timeline:

Event	Dates
Registration closes	As per the date of respective city zonal
Round 1 submission and Presentation	On the date of that city zonal
Results of Round 1	On the day of that city zonal
Final report submission deadline	1st December 2026
Finals	15-18th December

Evaluation, Prizes, and Recognition:

Evaluation Criteria

- **Creativity and Novelty**: How original and innovative is the proposed idea? How effectively does it introduce new smart technologies, IoT architectures, or mechanical approaches compared to existing solutions?
- **Functionality and Usability**: How well does the working prototype or simulation perform? Is the system reliable, accurate, and robust when operating under real-world environmental pressures (such as subterranean connectivity, marine corrosion, tidal shifts, or high-dust atmospheres)?
- **Cost and Market Value**: Is the solution economically viable and affordable? Does it present a sustainable financial model for long-term municipal, industrial, or community adoption?
- **Technical Implementation & Feasibility**: Is the solution technically feasible, repeatable, and resilient? How well does it account for operational challenges like power consumption, sensor calibration, maintenance, and severe weather conditions?
- **Regional Scalability**: Can the solution be easily scaled and adapted to diverse geographical, infrastructural, and regulatory landscapes across different cities and coastal hubs in India?
- **Environmental & Socioeconomic Impact**: What is the quantifiable impact on resource preservation, pollution mitigation (such as water conservation, AQI improvement, or marine plastic diversion), and public

health or community well-being?

- **Design & Ergonomics**: Is the system architecture, hardware layout, or software interface well-thought-out, user-centric, and optimized for operators, municipal authorities, or local communities?
- **Problem-Centric Approach (Bonus Points)**: Teams that demonstrate a deep, nuanced understanding of their chosen theme (Underground Water Management, Rising AQI, or Coastal Plastic Pollution in regions like Mumbai) will receive extra credit. This includes providing a detailed analysis of specific local pain points and showing how their solution directly targets those unique challenges rather than relying on a generic product.

General Rules

1. Teams must register online at **techfest.org**, receiving a Team ID for future reference.
2. Registration and Round 1 submissions are open until the date of respective City zonals.
3. The organizers and judges' decisions are final and binding. Techfest reserves the right to verify participant details.
4. Techfest, IIT Bombay, is not responsible for late or lost entries.
5. Solutions must be original and not protected by patents, copyrights, or publications.
6. The latest information will be available on **techfest.org**, with registered participants being notified of changes via email.
7. All technical, procedural, and competition-related queries must be submitted directly through the Techfest Competition Portal under the Doubt section. Participants are strictly requested to regularly check the portal for answers and updates
8. Official communication for important announcements will be through the satwik@techfest.org. Queries submitted via email will not be entertained

Certificate Policy

1. The top three teams per theme in Round 2 will receive a Certificate of Excellence.
2. All teams **attending** Round 2 at IIT Bombay will receive an **e-certificate**.

Prize Money

Prize money will be awarded to the top three teams per theme via NEFT by the latest May 2027

Prize Money Submission Process

- **Submission Channel:** Complete the prize money form **directly on the website** after the final round concludes. The winners will be able to view the form through their registered email ID
- **Deadline:** The form must be submitted by **December 31, 2026**.

Details Needed for the Website Form

When filling out the form on the website, make sure you have the following bank account details ready:

1. Account Holder's Name
2. Account Number
3. Bank Name and Branch Name
4. IFSC Code